Statistical Methods and Automated Softwate Testing

Research Proposal

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7th June 2017

Motivation

U$22.2 to U$59.5 billion!*

* [Tassey 2002].

Bug-Free Software?

- Formal verification
- Testing

A paper

Claessen, Koen and Hughes, John (2000). QuickCheck: A Lightweight Tool for Random Testing of Haskell programs. ICFP'00.

QuickCheck: From Functional Programming to the World

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An open source library

QuickCheck on Hackage (`hackage.haskell.org/package/QuickCheck`).

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Most Influential ICFP Paper Award 2010

"The techniques described in the paper have spawned a significant body of follow-on work in test case generation. They have also been adapted to other languages..." (from `www.sigplan.org/Awards/ICFP/`)

Adaptations

QuickCheck has been ported to C, C#, C++, Chicken Scheme, Erlang, F#, Go, Java, JavaScript, Julia, Node.js, Objective-C, OCaml, Perl, Prolog, Python, R, Racket, Ruby, Rust Scala, Scheme and Standard ML among others (from `en.wikipedia.org/wiki/QuickCheck` on 2017-04-04).

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Commercialisation

QuviQ (`www.quviq.com/`).

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The program works properly but the test pointed out a fail:

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False negative

There is a bug in the program but the test passed:

- *"Testing shows the presence, not the absence of bugs"* (Dijkstra 1969)
- Improvements from the statistical methods

References

Claessen, K. (2012). Shrinking and Showing Functions: (Functional Pearl). In: Proceedings of the 2012 Haskell Symposium (Haskell '12). ACM, pp. 73–80.

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– (2002). Testing Monadic Code with QuickCheck. In: Proceedings of the 2002 ACM SIGPLAN Workshop on Haskell (Haskell '02). ACM, pp. 65–77.

Tassey, G. (2002). The Economic Impacts of Inadequate Infrastructure for Software Testing. Tech. rep. National Institute of Standards and Technology. US Department of Commerce.